MindVault Architecture Specification: Distributed Cloud Storage and Sharding
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Section 1: Multi-Zone Replication and Quorum Management
The storage subsystem implements distributed active-passive chunk replication across three availability zones. Cross-zone consistency utilizes a quorum policy where write acknowledgments require W=2 confirmation out of N=3 replicas. Read operations enforce R=2 consensus before returning. Data blocks are partitioned using consistent hash rings to balance IO load.

Section 2: Chunk Hashing and Cryptographic Integrity
Every stored chunk is validated using a deterministic SHA-256 hash digest. Chunk address verification ensures that chunk content is not tampered with at rest. When paragraphs exceed targetTokens (700), recursive boundary decomposition splits on sentence punctuation with 120-token overlap, up to a hard ceiling of maxTokens (900).

Section 3: Raft Consensus Leases and Heartbeats
Leader election leases are renewed via periodic heartbeats set to 1500ms intervals. If the active leader fails to emit a heartbeat within this lease window, candidate nodes trigger an election protocol.
Data encryption at rest employs AES-256-GCM authenticated encryption for all persistent block stores.

Section 4: Subsystem Dependencies and Revocation Cascades
The Hybrid Retrieval Engine queries Vector Search Index directly. The Incident Health Daemon monitors replication lag across all shard leaders. Revocation of tenant master keys disables document decryption, the embedding pipeline, and context assembly.